

# Activities 2016 / 2017

Nora Juhasz

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### **Abstract**

This document describes the university classes, GSSI classes, summer schools and seminars attended in the 2016 / 2017 academic year.

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## 1 University of L'Aquila

### 1.1 Numerical Analysis 2 - Prof. Guglielmi

Euler method. Steepest descent gradient method. Conjugate directions method. Matlab.

### 1.2 High Performance Parallel Computing - Prof. Shcherbatyy, Prof. Pera

Basic Linux commands.

## 2 Gran Sasso Science Institute

### 2.1 Introduction to Finite Elements Method - Prof. Boffi

Finite differences, finite elements.

### 2.2 Decay property for hyperbolic systems with dissipation - Prof. Kawashima

The aim of the course is to study dissipative structure and decay property for hyperbolic systems with dissipation. Consider the following two equations.

$$u_t + Au_x = Bu_{xx} \quad (1)$$

$$u_t + Au_x - Lu = 0 \quad (2)$$

It is always assumed that the matrix  $A$  is symmetric. Throughout the course of this lecture different versions of these two equations are examined in terms of the assumptions that are made on matrices  $B$  and  $L$ , or what type of additional effects are involved. The simplest case is when one assumes  $B$  and  $L$  to be symmetric and nonnegative.

Equation 1 describes the symmetric diffusion process while symmetric relaxation is modeled by equation 2.

### 2.3 Advanced Topics on Numerical Methods - Prof. Guglielmi

A-stability, B-stability, Runge-Kutta.

### 2.4 Introduction to the Incompressible Navier-Stokes equation and the mathematical theory of Turbulence - Prof. Marcati

Euler and Navier-Stokes equations.

## 3 Seminars

### 3.1 PDEs Day, 21st March 2017

- 3.1.1 Viscous vortex rings with axial symmetry - Prof. Gallay
- 3.1.2 A two phase model with cross and self attractive interactions  
- Prof. Novaga
- 3.1.3 Traveling waves for collective movements on a network - Prof.  
Corli

## 4 Summer schools

## References

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